

Linear algebra: Exercise Chapter 8

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Autumn 2024

1. Determine which one of following matrix is diagonalizable and diagonalize them. Compute algebraic and geometric multiplicity for each eigenvalues.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 4 \\ 3 & 3 & 5 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 3 & -19 & 11 \\ 0 & 1 & 7 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Answer.

Matrix A: It has two eigenvalues $\lambda_1 = -1$ and $\lambda_2 = 3$.

$\lambda_1 = -1$:

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad M_g = M_a = 1.$$

$\lambda_2 = 3$:

$$v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad M_g = M_a = 1.$$

Matrix A is diagonalizable.

$$X = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}, \quad X^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Matrix B: It has two eigenvalues $\lambda_1 = -3$ (multiple eigenvalue) and $\lambda_2 = 5$.

$\lambda_1 = -3$:

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad M_g = M_a = 2.$$

$\lambda_2 = 5$:

$$v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad M_g = M_a = 1.$$

Matrix B is diagonalizable and $B = IBI$. Indeed,

$$X = X^{-1} = I, \quad \Lambda = \begin{bmatrix} -3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{bmatrix} = B.$$

Matrix C: It has three eigenvalues $\lambda_1 = 0$, $\lambda_2 = 1$, and $\lambda_3 = 8$.

$\lambda_1 = 0$:

$$v_1 = \begin{bmatrix} 3 \\ 2 \\ -3 \end{bmatrix} \quad M_g = M_a = 1.$$

$\lambda_2 = 1$:

$$v_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad M_g = M_a = 1.$$

$\lambda_2 = 8$:

$$v_2 = \begin{bmatrix} 1 \\ 6 \\ 7 \end{bmatrix} \quad M_g = M_a = 1.$$

Matrix C is diagonalizable.

$$X = \begin{bmatrix} 3 & 1 & 1 \\ 2 & -1 & 6 \\ -3 & 0 & 7 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 8 \end{bmatrix}, \quad X^{-1} = \frac{1}{56} \begin{bmatrix} 7 & 7 & -7 \\ 32 & -24 & 16 \\ 3 & 3 & 5 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 3 & 4 \\ 3 & 3 & 5 \end{bmatrix} = \frac{1}{56} \begin{bmatrix} 3 & 1 & 1 \\ 2 & -1 & 6 \\ -3 & 0 & 7 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 8 \end{bmatrix} \begin{bmatrix} \frac{1}{8} & \frac{1}{8} & -\frac{1}{8} \\ \frac{4}{7} & -\frac{3}{7} & \frac{2}{7} \\ \frac{3}{56} & \frac{3}{56} & \frac{5}{56} \end{bmatrix}.$$

Matrix D : It has two eigenvalues $\lambda_1 = 2$ and $\lambda_2 = 1$ (multiple eigenvalue).

$\lambda_1 = 2$:

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad M_g = M_a = 1.$$

$\lambda_2 = 1$:

$$v_2 = \begin{bmatrix} 1 \\ -3 \\ 0 \\ 0 \end{bmatrix} \quad M_g = 1, \quad M_a = 3.$$

Matrix D is nondiagonalizable.

2. Consider the matrix A in question 1. Compute A^5 .

Answer.

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Thus,

$$A^5 = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} (-1)^5 & 0 \\ 0 & 3^5 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 243 \\ 1 & 243 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 122 & 121 \\ 122 & 121 \end{bmatrix}.$$

3. Suppose A is a diagonalizable matrix with $A = X\Lambda X^{-1}$. Diagonalize $A + 2I$.

Answer. If λ is an eigenvalue of A , then $\lambda + 2$ is an eigenvalue of $A + 2I$. Also, the eigenvectors are the same. So, $A + 2I = X(\Lambda + 2I)X^{-1}$.

4. Let A, B and C be $n \times n$ matrices. Prove the following statements:

- If A is similar to B and B is similar to C , then A is similar to C .
- If A is similar to B , then A^k is similar to B^k , for each $k \in \mathbb{N}$.
- If A and B are similar matrices, then their determinants are the same.

Answer.

- Matrix A is similar to B and B is similar to C , so there exist invertible matrices P and Q such that $A = PBP^{-1}$ and $B = QCQ^{-1}$. Set $X = PQ$. Then,

$$A = PQCQ^{-1}P^{-1} = PQC(PQ)^{-1} = XCX^{-1}.$$

Therefore, A is similar to C .

- Matrix A is similar to B , so there exists invertible matrix P such that $A = PBP^{-1}$.

$$PB^kP^{-1} = PBP^{-1}PBP^{-1}PBP^{-1} \dots PBP^{-1} = AA \dots A = A^k.$$

Therefore, A^k is similar to B^k .

- Matrix A is similar to B , so there exists invertible matrix P such that $A = PBP^{-1}$. Then,

$$\det(A) = \det(PBP^{-1}) = \det(P)\det(B)\det(P^{-1}) = \det(B).$$

5. True or false: If the columns of matrix X are the eigenvectors of an $n \times n$ matrix A and they are linearly independent, then

- (a) A is invertible;
- (b) A is diagonalizable;
- (c) X is invertible;
- (d) X is diagonalizable.

Answer. Since the columns of matrix X are linearly independent, so X is full rank and it is invertible. Also, matrix A has n linearly independent eigenvector, so it is diagonalizable. Thus, items (b) and (c) are true. But, items (a) and (d) are not true; consider

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix},$$

where $\det(A) = 0$ and $X = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

6. True or false: If the only eigenvectors of A are multiples of $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$, then

- (a) A is not invertible;
- (b) A has a repeated eigenvalue;
- (c) A is nondiagonalizable.

Answer. Matrix A has one repeated eigenvalue with one linearly independent eigenvector $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ and so, it is nondiagonalizable. Therefore, the items (b) and (c) are true. But, part (a) is false, because it is possible that A is invertible. Consider:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$